Benchmarking Projected Generator Coordinate Method with One-Quasiparticle Excitation for Odd-A Nuclear β Decay

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Introduction: The Gamow-Teller Quenching Puzzle

- **The Puzzle**: Standard spherical shell model calculations systematically overestimate the Gamow-Teller (GT) strength. To match experimental decay rates, an empirical quenching factor $q \approx 0.75$ must be introduced to scale the axial-vector coupling g_A.
- **Possible Mechanisms**:

1. Operator Renormalization (Chiral Two-Body Currents)

Choral Effective Field Theory (χEFT) introduces two-body weak currents (TBC) which modify the transition operator and quench the effective g_A at the operator level.

2. Wavefunction Correlations (Many-Body Effects)

Multi-shell configuration mixing, intrinsic deformation, and complex many-body correlations.

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Introduction: The Quenching Puzzle

- Using SRG-evolved NN and 3N interactions, the NCSM, the coupled-cluster method, and the VS-IMSRG are used to compute nuclei in different regions.Two-body currents contribute significantly to the quenching.
  *P. Gysbers et al., Nature Phys. **15**, 428 (2019)*
- Applying shell-model MBPT to construct an effective GT operator from χEFT for pf-shell nuclei, the chiral expansion of the electroweak currents and the many-body renormalizations, share equal merit in providing a noteworthy reproduction of data.
  *L. Coraggio et al., Phys. Rev. C **109**, 014301 (2024)*
- Employing the projected shell model with chiral two-body currents for ^{76}Ge , many-body correlations (deformation, cross-shell) dominate GT quenching while two-body currents contribute only 5–15%, though the Hamiltonian uses a separable force not consistently derived from χEFT.
  H. Zhou, L.-J. Wang, Y. Sun, arXiv:2509.24542 (2025)

Introduction: The Quenching Puzzle

- Combining ab initio VS-IMSRG with chiral NN and 3N interactions and two-body currents for $N = 50$ neutron-rich isotones ,The available experimental half-lives are well described once 2B cur rents are included. FF con tributions suggests that they are smaller below Z = 27.Future work should include more detailed stud ies of the EFT interaction uncertainties and a consistent inclusion of FF contributions with 2B currents as well

  Z. Li, H. Miyagi, A. Schwenk et al., Phys. Rev. Lett. ***136****, 182501 (2026)*
- Using self-consistent RPA + PVC with chiral two-body currents for three doubly magic nuclei, 56Ni, 100Sn, and 132Sn. PVC and two-body currents jointly yield $q \approx 0.73$–0.80 and reproduce experimental GT strengths .Among the several contributions of chiral TBC, the meson-exchange terms give the major contribution to the quenching, to the choice of LEC parameters, while the momentum-dependent term has a negligible contribution. *B.-L. Wang et al., arXiv:2605.21218 (2026)*

Method

PSM

- First systematic β-decay (GT) theory in the PSM; derived 1qp/3qp matrix elements and showed multi-qp mixing is essential, .Z.-C. Gao, Y. Sun, Y.-S. Chen, Chin. Phys. C **30**(S2), 65 (2006)
- Extended the PSM truncation from 1qp/3qp up to 4-qp/5-qp using a Pfaffian, computing bound-state β half-lives (GT + first-forbidden).J.-W. Ran, L.-J. Wang *et al.*, arXiv:2606.08963 (2026)
- Identified nuclear deformation and cross-shell many-body correlations — not chiral two-body currents — as the main origin of Gamow-Teller quenching (^{76}Ge).L.-J. Wang *et al.*, arXiv:2509.24542 (2025)
- PSM study of β-delayed γ emission and mirror asymmetry in ^{26}P.Y. Sun *et al.*, Phys. Rev. Lett. **127** (2021)

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Method

QRPA (blocked HFB $\approx$ 1qp for odd-A)

- Axially deformed relativistic pnQRPA with point-coupling interactions.A. Ravlić *et al.*, PRC **110**, 024323 (2024)
- Added exact angular-momentum restoration to deformed pnQRPA, shortening $^{62-68}$Fe half-lives by up to 60%.A. Ravlić *et al.*, arXiv:2510.16313 (2025)
- Spherical covariant QRPA global β-decay half-lives; systematic extraction of the effective $g_A \approx 0.6$–0.8.T. Marketin *et al.*, PRC **93**, 014322 (2016); J. Suhonen, Front. Phys. **5**, 55 (2017)

GCM + PNAMP

- Gogny D1S HFB with PNP + AMP + GCM for $0\nu\beta\beta$ matrix elements; odd-A single-β (1qp/3qp) still unexplored.T. R. Rodríguez, J. L. Egido, arXiv:1008.5260 (2010)
- *Ab initio* IMSRG Hamiltonian combined with GCM for the ^{48}Ca $0\nu\beta\beta$ matrix element.H. Hergert, R. Stroberg *et al.*, arXiv:1908.05424 (2019)

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The sd-shell Hamiltonian

$$H = \sum_{p} \varepsilon_p \, c_p^\dagger c_p - \frac{1}{4} \sum_{pqrsJM} \mathcal{V}^J_{pqrs} \, (-1)^{J+M} [c_p^\dagger c_q^\dagger]_{JM} \, [\widetilde{c}_r \, \widetilde{c}_s]_{J-M}$$

USDB interaction: defined within the sd shell on top of an inert ^{16}O core

Orbitals: $0d_{5/2}$, $1s_{1/2}$, $0d_{3/2}$ for protons and neutrons

PGCM wave function for low-lying states

The α-th state with quantum numbers $NZJM\pi$ is expanded as

$$\boxed{|\Psi_{\alpha}^{NZJM\pi}\rangle = \sum_{c} f_{\alpha c}^{NZJM\pi}\, \hat{P}_{MK}^{J}\, \hat{P}^{N} \hat{P}^{Z}\, |\Phi_{\kappa}(\mathbf{q})\rangle}$$

Collective index

$$c = \{K,\ \kappa,\ \mathbf{q}\}$$

- K — the component of angular momentum along the intrinsic axis z
- κ — labels the quasiparticle configuration
- $\mathbf{q}$ — shape (deformation) parameters

Symmetry assumption

Unless otherwise stated, the deformed configurations are assumed to have **axial symmetry**.

Restoration operators

Angular-momentum projector

$$\hat{P}^{J}_{MK} = \frac{2J+1}{8\pi^2} \int d\Omega \, D^{J*}_{MK}(\Omega) \, \hat{R}(\Omega)$$

- $\Omega = (\phi, \theta, \psi)$ Euler angles
- $D^{J}_{MK} = \langle JM|\hat{R}(\Omega)|JK\rangle$
- $\hat{R} = e^{i\phi\hat{J}_z} e^{i\theta\hat{J}_y} e^{i\psi\hat{J}_z}$

Particle-number projector

$$\hat{P}^{N_\tau} = \frac{1}{2\pi} \int_0^{2\pi} d\varphi_\tau \, e^{i\varphi_\tau(\hat{N}_\tau - N_\tau)}$$

- $\tau = n, p$ (applied separately)
- $\hat{N} = \sum_k a^\dagger_k a_k$
- Extracts the component with exact N_τ

K is the projection of the angular momentum on the intrinsic symmetry axis; M is its laboratory-frame counterpart.

Mean-field configurations

Odd-mass nucleus

A **one-quasiparticle** excitation on an even-parity vacuum:

$$|\Phi_{\kappa}^{(\mathrm{OA})}(\mathbf{q})\rangle = \alpha_{\kappa}^{\dagger}\,|\Phi_{(\kappa)}(\mathbf{q})\rangle, \qquad \alpha_{\kappa}|\Phi_{(\kappa)}\rangle = 0$$

The qp operators follow from the Bogoliubov transformation

$$\begin{pmatrix} \alpha \\ \alpha^{\dagger} \end{pmatrix} = \begin{pmatrix} U^{\dagger} & V^{\dagger} \\ V^{T} & U^{T} \end{pmatrix} \begin{pmatrix} a \\ a^{\dagger} \end{pmatrix},$$

where U, V are obtained from the self-consistent VAPNP + HFB calculation (variation after particle-number projection).

Self-consistent one-quasiparticle blocking

The k-th odd-parity 1qp state is obtained **self-consistently** by exchanging the k-th column of U, V in the HFB wave function:

$$\boxed{(U_{pk}, V_{pk}) \longleftrightarrow (V^*_{pk}, U^*_{pk})}$$

with $p = (\tau\, n\ell j\Omega)_p$ a spherical HO basis label and k the quasiparticle label.

Good quantum number

- $K = \Omega_p$ (projection of j_p on z)
- parity $\pi = (-1)^{\ell_p}$

VAPNP condition

$$\delta \frac{\langle \Phi^{(\mathrm{OA})}_\kappa | \hat{H}\hat{P}^N\hat{P}^Z | \Phi^{(\mathrm{OA})}_\kappa \rangle}{\langle \Phi^{(\mathrm{OA})}_\kappa | \hat{P}^N\hat{P}^Z | \Phi^{(\mathrm{OA})}_\kappa \rangle} = 0$$

Collective coordinate: $\beta_2 = \dfrac{4\pi}{3AR^2}\langle\Phi|r^2 Y_{20}|\Phi\rangle, \quad R = 1.2A^{1/3}$ fm.

Quasiparticle wave function

1)

$$\mathcal{H} = \mathcal{W}_1^\dagger \mathcal{H}_{\mathrm{HFB}} \mathcal{W}_1 = \begin{pmatrix} H^{11} & 0 \\ 0 & -H^{11*} \end{pmatrix}$$

$$\mathcal{W}_1 = \begin{pmatrix} U_0 & V_0^* \\ V_0 & U_0^* \end{pmatrix}$$

2)

$$\mathcal{W}_2 = \begin{pmatrix} X & Y^* \\ Y & X^* \end{pmatrix}$$

obtained by solving the $2M \times 2M$ quadratic HFB equation for H^{11}.

$$U = U_0 X + V_0 Y, \qquad V = V_0 X + U_0 Y.$$

Check for ^{20}Ne (USDB, sd space)

Bogoliubov orthonormality

$$\|X^TX + Y^TY - I\| = 1.78 \times 10^{-15}, \qquad \|X^TY + Y^TX\| = 4.82 \times 10^{-16}$$

$\mathrm{Tr}(VV^T) = \sum_{m,k} V_{mk}^2 =$ **4.0002**

$\mathrm{Tr}(UU^T) = \sum_{m,k} U_{mk}^2 =$ **19.9998**

sum = **24.0000**

- $\mathrm{Tr}(VV^T)$ = expected particle number in the qp vacuum.
- 24 = dimension of the full sd model space ($0d_{5/2}, 1s_{1/2}, 0d_{3/2}$);

Bogoliubov unitarity check

The matrices U and V define a unitary Bogoliubov transformation for the quasiparticle wave function.

$$
\begin{aligned}
\|U^T U + V^T V - I\|_F &= 5.46 \times 10^{-13} \\
\|U^T V + V^T U\|_F &= 6.68 \times 10^{-15} \\
\|U U^T + V V^T - I\|_F &= 5.46 \times 10^{-13} \\
\|U V^T + V U^T\|_F &= 1.78 \times 10^{-14}
\end{aligned}
$$

Evaluation of $\langle J_z \rangle$ in the HO Basis

The density matrix in the harmonic-oscillator (HO) basis is

$$\rho_{mn} = \sum_k V^*_{mk} V_{nk}, \tag{1}$$

where m, n label HO single-particle states and k labels quasiparticle states. Since the HO basis states are eigenstates of J_z,

$$(J_z)_{mn} = K_m \delta_{mn} \tag{2}$$

$$\begin{aligned} \langle J_z \rangle = \mathrm{Tr}(\rho J_z) &= \sum_{m,n} \rho_{mn} (J_z)_{nm} = \sum_m K_m \rho_{mm} \\ &= \sum_m K_m \sum_k |V_{mk}|^2. \end{aligned} \tag{3}$$

For an axially symmetric state with blocked quasiparticles

$$K_{\mathrm{tot}} = K_1 + K_2 + K_3. \tag{4}$$

Consistency Check of the HFB Density Matrix

Comparing the density matrices before and after the quasiparticle-basis transformation,

$$\rho^{(0)} = V_0 V_0^\dagger, \qquad \rho = V V^\dagger, \tag{5}$$

we define the maximum deviation

$$\epsilon_\rho = \max_{m,n} \left| \rho_{mn} - \rho_{mn}^{(0)} \right|. \tag{6}$$

Numerical result:

$$\epsilon_\rho = 3.546 \times 10^{-3}, \tag{7}$$

(8)

If $Y = 0$ and X is unitary, with

$$U = U_0 X, \qquad V = V_0 X, \tag{9}$$

then

$$V V^\dagger = V_0 X X^\dagger V_0^\dagger = V_0 V_0^\dagger, \tag{10}$$

Numerical Diagnostics

Diagnostic	Numerical result	Interpretation
$\max_{ij} \lvert Y_{ij} \rvert$	7.1530×10^{-4}	Small maximum element of Y
$\max_{ij} \lvert (XX^T - I)_{ij} \rvert$	8.3758×10^{-7}	X is nearly orthogonal
$\lVert Y_{\text{mat}} \rVert_F$	2.491×10^{-3}	Small Frobenius norm of reconstructed Y
$\lVert X_{\text{mat}} \rVert_F$	4.899	Overall magnitude of reconstructed X

Bogoliubov transformation

$$U = U_0 X + V_0 Y, \qquad V = V_0 X + U_0 Y.$$

Preliminary assessment

- X is nearly orthogonal, with a maximum deviation of approximately 10^{-6}.
- The reported magnitudes of Y are small.
- These checks alone do not guarantee that the transformation and quasiparticle blocking are fully consistent.

Calculation of the Projected J_z Kernel

Angular-momentum projection operator

$$\hat{P}^J_{MK} = \frac{2J+1}{8\pi^2} \int d\Omega\, D^{J*}_{MK}(\Omega)\hat{R}(\Omega).$$

Projected transition expectation value

$$\mathcal{J}^J_z = \frac{\langle\Phi_f|\hat{J}_z\hat{P}^J_{MK}|\Phi_i\rangle}{\langle\Phi_f|\hat{P}^J_{MK}|\Phi_i\rangle} = \frac{\int d\Omega\, D^{J*}_{MK}(\Omega)\langle\Phi_f|\hat{J}_z\hat{R}(\Omega)|\Phi_i\rangle}{\int d\Omega\, D^{J*}_{MK}(\Omega)\langle\Phi_f|\hat{R}(\Omega)|\Phi_i\rangle}.$$

Transition-density representation

$$\rho_{ji}(\Omega) = \frac{\langle\Phi_f|c^\dagger_i c_j\hat{R}(\Omega)|\Phi_i\rangle}{\langle\Phi_f|\hat{R}(\Omega)|\Phi_i\rangle}.$$

$$\langle\Phi_f|\hat{J}_z\hat{R}(\Omega)|\Phi_i\rangle = \langle\Phi_f|\hat{R}(\Omega)|\Phi_i\rangle\,\mathrm{Tr}[j_z\rho(\Omega)]\,,\mathrm{Tr}(j_z\rho) = \sum_{ij}(j_z)_{ij}\rho_{ji}.$$

Eigenvalue check Since $[\hat{J}_z, \hat{P}^J_{MK}] = 0$ and $\hat{J}_z\hat{P}^J_{MK} = M\hat{P}^J_{MK}$,

$$\boxed{\mathcal{J}^J_z = M}$$

Calculation of the J_z Transition Kernel

1. Theoretical formula

For the one-body angular momentum operator

$$\hat{J}_z = \sum_{ij} (j_z)_{ij} c_i^\dagger c_j,$$

the normalized transition expectation value is

$$\boxed{\mathcal{J}_z^J = \frac{\langle \Phi_f | \hat{J}_z \hat{P}^J_{MK} | \Phi_i \rangle}{\langle \Phi_f | \hat{P}^J_{MK} | \Phi_i \rangle} = \frac{\mathrm{Tr}\left(j_z \rho^J_{fi}\right)}{1}}$$

where ρ is the transition density matrix, assuming $\rho_{ji} = \langle c_i^\dagger c_j \rangle$.

2. Fortran implementation

```
^^I^^I^^Icall ZGEMM('N','N',NDIM,NDIM,NDIM,zone, &
^^I^^I^^IcME1B%ZJzME,NDIM,ZRO,NDIM, &
^^I^^I^^Izzero,zAUX,NDIM)
^^I^^I^^I
^^I^^I^^IzJZ_MV = zzero
^^I^^I^^Ido ii = 1, NDIM
^^I^^I^^IzJZ_MV = zJZ_MV + zAUX(ii,ii)
^^I^^I^^Ienddo
^^I^^I
```

This evaluates